

Omar Rayyan

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EDUCATION

University of California, Los Angeles (UCLA)

September 2025 - Present

PhD Student (Advised by Prof. Yuchen Cui)

New York University (NYU)

August 2021 - May 2025

Double Major in Electrical Engineering & Computer Science

GPA: 4.0/4.0 (Summa Cum Laude)

EXPERIENCES

GRAIL (NYU Courant) — Advised by Prof. Lerrel Pinto

January 2024 - Present

Research Assistant

New York, US

- Collaborating with the AI2 PRIOR team on a THOR-based project.
- Developing MuJoCo-based simulation evaluation environments for RUM utility models.
- Led the UntidyBot project, a mobile manipulation framework designed to perform logical, sequential actions in diverse environments beyond standard pick-and-place tasks.
- Open-sourced MuJoCo AR, a plugin enabling real-time integration of ARKit data from iOS devices to control MuJoCo frames and real robots. Installed 20k times since release.

PAIR Lab (Georgia Tech) — Advised by Prof. Animesh Garg

July 2024 - Present

Research Assistant

Atlanta, US

- Scaling data collection for imitation learning using Isaac-Sim and MuJoCo, allowing users to teleoperate vectorized, cloud-hosted IsaacSim environment. Our COBALT project is currently in-submission.
- Designed and implemented custom environments and robotic manipulation tasks in simulation.

CAIR (NYU Abu Dhabi) — Advised by Prof. Anthony Tzes

September 2022 - July 2025

Research Assistant

Abu Dhabi, UAE

- Worked on MV-UMI a scalable multi-view interface for cross-embodiment learning. It allows UMI to use a third-person camera input while mitigating domain shifts. It is currently in submission.
- Formalized the Remote Center of Motion (RCM) constraint as an algorithmically optimizable task function for general-purpose robotic manipulators, diverging from the mechanically constrained approach of Da Vinci systems. I published and presented this work at the International Conference on Advanced Robotics (ICAR 23')

Google

Software Engineering Intern

Jul 2023 - Sep 2023

Munich, Germany

- Developed a tool to automatically evaluate changes in resource configurations before their integration into the Google Cloud reliability pipeline
- My project earned an award by the team for creating a direct impact on their workflow during my internship

Applied Interactive Multimedia (AIM) Lab — NYUAD

February 2022 - Sep 2022

Student Researcher

Abu Dhabi, UAE

- Worked on KATIB by developing rehabilitation tasks using kinesthetic feedback via under-screen magnets to aid post-stroke recovery, in collaboration with Cleveland Clinic Abu Dhabi physicians.

SELECT PUBLICATIONS AND SOFTWARE

COBALT: Crowdsourcing Robot Learning via Cloud-Based Teleoperation	Website
<i>Agarwal, A., Gandh, A., Collins, J., Rayyan, O., Sarsawat, A., Koushik, R., .. & Garg, A. In Submission</i>	
MV-UMI: A Scalable Multi-View Interface for Cross-Embodiment Learning	Website
<i>Rayyan, O., Abanes, J., Hafez, M., Abu-Daka F., & Tzes A. In Submission</i>	
RCM-Constrained Manipulator Trajectory Tracking using Differential Kinematics	
<i>Rayyan, O., Gonçalves, V., Evangeliou, N., & Tzes</i> ICAR 2023	
MuJoCo AR	
<i>ARKit teleoperation for MuJoCo and the real world</i> Repository	
UntidyBot	
<i>ok-robot, beyond pick and place</i> Repository	
Past Code	
<i>App with over a million usages that helps students find supporting papers using their phone.</i> App Store	

AWARDS AND HONORS

IEEE Best Project Prize	May 2025
Awarded the best project prize by IEEE for my bachelor thesis across the UAE.	
Comma.ai Controls Competition	Aug 2024
1st place in comma.ai's controls challenge by training a DDPG reinforcement learning agent to steer a car	
IEEE Xtreme Coding Competition Podiums	2022/23 & 2023/24
Led a team to podium twice across the region at IEEE's programming competition	
Alfred H. Bloom Scholar	Jun 2024
Granted annually to one student for excellence in experiential learning - (Letter)	
Deep Learning Graduate Course Competition	May 2024
Led an undergraduate team to podium in the graduate-level deep learning course competition at NYU	
StartAD Innovention Competition Finalist	May 2022
Founded LEBSI, a social marketplace, which reached the App Store top charts.	
Cambridge Outstanding UK Learner Award	2021
Achieved the highest mark in country and region in my Physics A-Level, and across 7 IGCSEs	

EXTRACURRICULARS

Competitive Cubing (Rubik's Cube)	January 2017 - Present
- Set the official 4x4 national record (average and single) at an official WCA cubing competition - Competed in cubing events in Amman, Manama, New York, and Abu Dhabi - Led a Rubik's Cube speed-solving workshop at ZINC (Zain Innovation Campus)	